

Assignment 1

COS721

2026

Weight: 30% of your semester mark.

Due: Monday, 14 September 2026.

Upload to ClickUP.

Note that this is an individual assignment; you are only allowed to submit work that you have done on your own.

1 Introduction

For this assignment you are expected to propose and evaluate toolmarks to help classify PDF documents, with toolmarks understood as used in my paper on the topic [1]. In this assignment we will refer to this paper as the *underlying paper* or, where the context is clear, simply as *the paper*.

This introduction describes the context and concepts. While they should have a significant impact on the report you submit, it is possible that your report will not mention some of these issues at all. The expectations about the content of your assignment are provided in section 3 below. Also note that some technical expectations (such as maximum length and the font size to use are specified towards the end of this document).

Let us refer to the prefixes used in the paper as *loci*. Hence `stream-*` is a locus, as is `length-*`. The toolmarks you propose for this assignment should centre around two loci not explored in the paper — neither literally, nor in spirit. The first prohibition means that if you want to use the `pagetype-*` locus discussed in an announcement, you cannot call it the `page-*` locus, because the `page-*` locus is already used in the paper. The second prohibition means that you cannot propose, say, a `streamterm-*` locus that reflects how the `stream` keyword is terminated, because that is exactly what the `stream-*` locus is used for.

Some obvious strategies to find new loci would be to consider (a) line terminations (or, in general, what follows some keyword); (b) the sequence in which tokens appear (such as `seq-*` and `pagetype-*` from the announcement); (c) checks whether a token is present or absent (such as `xref-*` and `objstm-*`; and (d) usage conventions (such as `length-*`). You are encouraged to think beyond these four strategies. Some of these strategies also come with caveats: We noted that line terminations quickly become boring; while they are not

banned, your mark for a line-termination locus is likely to be significantly lower than a locus that just mirrors some existing locus. However, it may be a good idea to start with such a simple locus, incorporate it into the ‘system’, and confirm that it does not really contribute. This will confirm that you understand the operation of the ‘system’. Note that such a confirmation will not directly contribute to your marks for the assignment; the confirmation is intended as a self-check to confirm to you that you do understand the way the ‘system’ works.

Note that it is expected that you will explore some locus and initially find it interesting, but later determine that it is not a good choice. It seems like a good idea to prepare a list of potential loci so that picking another one is simple if a chosen one does not meet your expectations. In fact, trying a few and then writing up your best two loci seems worthwhile.

2 Initial suitability checks

Consider the `length-*` locus. It consists of `length-absent`, `length-direct` and `length-indirect` marks, as well as the complements of the latter two. Suppose it rather only consisted of `length-absent` and `length-present`. Then one document would have been associated with the former mark, while 999 would have been associated with the latter mark. In this form it clearly would not have had discriminatory power. Any mark that is too prevalent does not shed much insight. A mark with low prevalence may, however, identify some unique aspect of a scarce tool that behaves in some unusual manner. In this particular case the document without a `/Length is 6HTC5`, which clearly is not an ‘unusual’ PDF, but not a PDF document at all. Another way to think about the `length-*` locus is that documents with `/Length` tokens will manifest as `length-direct` and/or `length-indirect`. Would flagging them as `length-present` not be redundant?

A second common sense check is the nature of the locus you are considering. You may, for example, see `/DCTDecode` in a number of documents — almost 200, if my quick check is correct. However, when you look at the PDF reference manuals, you will find that `/DCTDecode` is closely linked to JPEG compression. Hence, such a mark *may* differentiate between documents that embed JPEG images, and those that do not. We are interested in marks left by the creation tool, rather than marks that depend on the content of the document.

A third check was discussed in the announcement. Ideally marks at a locus should be consistent for a given tool, unless it adds discriminatory power.

Most of the preceding remarks deal with making appropriate choices about the two loci you will propose. In many cases, these issues will not feature in your report at all.

3 Your assignment

As noted, your assignment deals with proposing and assessing two loci and their associated marks. This section splits the task into various subtasks. Marks will be awarded for the subtasks; hence it is suggested that you use this section as a template for your submission.

3.1 The loci and toolmarks

Briefly name and describe the loci you selected, as well as their associated marks. A table listing the prevalence of the various marks will be useful. Ideally the table will suggest that some variety is present. Consider table 3 in the underlying paper. The observation that **stream-cr-lf** and **stream-lf**, respectively, represent about 66% and 33% of the dataset may be reason for optimism. At this stage **stream-cr-other** and **stream-absent** seem less useful, but are necessary to cover the entire dataset. Consider table 10. Even the (almost) 80/20 split turned out to be useful. Note that this check is just a rule-of-thumb check: some variety is necessary but not sufficient to ensure utility.

3.2 Consistency of marks within tools

As explained in an announcement, this task essentially requires you to perform

```
minus class mark | wc -l
```

where **class** is iterated over the ten or so different classes in the paper, and **mark** is iterated over the marks associated with your proposed loci. If matters work perfectly, all marks associated with a locus will lead to an empty set, except for one mark that will yield all the documents in the class used in the calculation. Suppose your locus is called **1-*** with associated marks **1-a**, **1-b** and **1-c**. Then you may find that class 1 in the paper is consistently associated with, say, **1-b**. Then it would be simple to report that **1-b** is characteristic of class 1 (and, hence, also classes 2 and 3). In fact, a simple table with rows labelled **1-a**, **1-b** and **1-c** may be a good starting point. The classes associated with each of these marks may then be provided in the second column, next to the appropriate row label.

If the marks are inconsistent across multiple classes, it is reason for concern. One option is to explain these inconsistencies. Another option is to pick another locus, which may lead to simpler outcomes.

Suppose the marks are mostly consistent. Is it possible to explain the inconsistencies? This *may* be a great opportunity to demonstrate insight; demonstrations of insight tend to lead to higher marks.

Suppose you find inconsistencies for, say, class 1. Recall that classes 1, 2 and 3 started from different tools. If your marks differentiate between tools 1, 2 and 3 (or between two of them), it may also be very useful because the discriminatory power of the ‘system’ will be increased. In fact, it is quite easy to differentiate between some of the ‘compound’ classes

to better identify the (different) tools that were ‘merged’ into one bigger ‘compound’ class. It may be worth considering this when you start looking for loci.

Ultimately, this part of your assignment may report the classes for which your toolmarks are consistent. It may additionally reflect on the reasons why inconsistencies are present. Hopefully any inconsistencies will not diminish discriminatory power. This part *may* be a good place for you to demonstrate insight.

3.3 Integration with other toolmarks

Use the environment provided for the underlying paper. Add your toolmarks to the **marks** directory. Regenerate the mark vectors associated with each of the 1000 documents. Regenerate the feature vectors and induced classes. Very little (if any) of this needs to be discussed in your report (unless something worth discussing emerges, which seems unlikely).

Now compare the (new) induced classes with the versions that existed before you added your toolmarks to the environment. Discuss any differences. (As described, the process may guide you to a situation where the new induced classes are identical to the old ones; such an observation would be fine.)

Next, compare the new feature vectors to the versions that existed initially. If your toolmarks do not appear in (at least, some of) the new vectors, something is wrong, and you will have to return to earlier stages of the process — perhaps even to the beginning.

In which feature vectors do your toolmarks appear? If only in some of them, is it worth discussing? In general, these comments would probably be brief.

Compare your loci with other loci in the feature vectors? Can you find any loci that correlate exactly with your loci? As an example, suppose every time your **l-a** appears in a feature vector, **stream-lf** is also present (and vice versa). Also, whenever your **l-b** appears, **stream-cr-lf** is present (and vice versa). In addition, every time your **l-c** is present, **stream-cr-other** is also present (and vice versa). Then there is some specific correlation between **l-*** and **stream-***. Is this correlation explicable, or is it spurious?

A looser correlation may exist where, say **l-b** always coincides with **stream-cr-lf**, but other marks associated with **l-*** and **stream-*** do not match. If you do not observe the strict correlation from the previous paragraph, discuss your observations about any less strict correlation.

If you cannot find any correlations between your loci and others, discuss any correlations that exist within the provided toolmarks. Reflect on your observations. Also attempt to explain why such correlations exist, but you were unable to find any that involve your loci. (If you only find ‘boring’ partial correlations for your loci, you are free to mention them and then rather discuss more interesting correlations in the provided toolmarks.)

Note that various (abstract) approaches exist for finding the correlations discussed here. One approach would be to construct new sets, with each set named after a toolmark. As elements, each new set contains the classes in which the toolmark appears in the feature vector. One of the scripts provided with the paper performs exactly such ‘flipping’. Then calculate the MD5 hashes of all these sets and sort according to the digests. It will be simple to see which toolmarks appear in the same feature vectors. As a special case, note

that some toolmarks do not appear in any feature vectors and their sets will be empty here. Do not infer too much from these empty sets that all map to the same digest.

For the sake of illustration, I did exactly what I described in the previous paragraph and created the sets in a directory called `seenin`. Then I executed the command `md5sum seenin/endstream-* seenin/endobj-* | sort`. This produced the following output (to which I added some blank lines for the sake of readability):

```
07c1b603cdf3b3272f951c5f8741495a  seenin/endobj-lf
07c1b603cdf3b3272f951c5f8741495a  seenin/endstream-lf

16a4d9885261b3b4a346c8a88890a675  seenin/endobj-NO-cr-lf
16a4d9885261b3b4a346c8a88890a675  seenin/endstream-NO-cr-lf

4b216bbe4bf1d8dc1148b96c8ef1752a  seenin/endobj-NO-lf
4b216bbe4bf1d8dc1148b96c8ef1752a  seenin/endstream-NO-lf

93b3dd5648c9deccf5aed6980c0eed09  seenin/endobj-cr-lf
93b3dd5648c9deccf5aed6980c0eed09  seenin/endstream-cr-lf

acfe70e01a87ff686d6f6a7593d45dce  seenin/endobj-cr-other
acfe70e01a87ff686d6f6a7593d45dce  seenin/endstream-cr-other

aec1e69b314a1a5d0bd685446f9cfb25  seenin/endobj-NO-cr-other
aec1e69b314a1a5d0bd685446f9cfb25  seenin/endstream-NO-cr-other

d41d8cd98f00b204e9800998ecf8427e  seenin/endobj-absent
d41d8cd98f00b204e9800998ecf8427e  seenin/endobj-other
d41d8cd98f00b204e9800998ecf8427e  seenin/endobj-space-cr
d41d8cd98f00b204e9800998ecf8427e  seenin/endobj-space-lf
d41d8cd98f00b204e9800998ecf8427e  seenin/endobj-space-other
d41d8cd98f00b204e9800998ecf8427e  seenin/endstream-space-cr
d41d8cd98f00b204e9800998ecf8427e  seenin/endstream-space-cr-lf
d41d8cd98f00b204e9800998ecf8427e  seenin/endstream-space-lf
d41d8cd98f00b204e9800998ecf8427e  seenin/endstream-space-other
```

The block at the bottom (d41...27e) are all empty. Note the perfect correlation between how `endstream` and `endobj` are terminated with a CR, LF or CR/LF (as well as not terminated by these characters). Hence, strict correlation exists in the provided data. However, this was expected for these toolmarks and discussing this would not provide much new insight.

3.4 Reflection

In the conclusion to your submission give a step back and reflect on your observations made while working on your assignment. You may have seen many opportunities that could have been selected as loci, or you may have struggled to even have found the two discussed in your report. The first loci you picked, may have withstood the challenges posed by the assignment, or you may have determined that they were poor choices and had to select new ones. At this stage you should have some insight into the ease with which new loci (and associated toolmarks) may be selected, and may have developed some insights into pitfalls that have to be avoided. You may be impressed by the manner in which classification works and/or you may have developed some significant concerns. (If your first picks just worked, it may be appropriate to consider two additional loci, just to confirm that selection is as simple as you probably deem it.)

Use this final section of your report to reflect on the usability of toolmarks (as encountered in this assignment) to classify PDF documents. Illustrate points you make with specific examples as far as possible.

4 Assessment

You already know that your submission will be awarded a mark out of 30. The four sections of your assignment will, respectively, be marked out of 5, 5, 10 and 10 marks. (In some cases the rubric may be adjusted; if your second section is, for example, weighted heavier than your final section, the second section may be marked out of 10, and the final one out of 5.) For each of the four parts of your answer, an impression mark will be assigned. This impression mark will be based on two factors: How hard or simple to execute was the task you set for yourself to execute, and how well did you execute it? Sometimes less is more: You may have to discard some really impressive goals that are too hard to report on: If your task is (almost) impossible and you fail to deliver, you will not do well. On the other hand, it is possible to select an extremely simple approach where the simplicity (and naivety) of the approach does not warrant a high mark. To get the best possible mark, plan to finish your assignment ahead of schedule and carefully reread it before submission. That will enable you to better judge whether the (reasonable) task that you have chosen, has been executed as well as possible.

5 Submission

1. Submit your final answer to the upload slot on clickUP. (You are allowed to submit as many reports and as often as you want; your final submission will be assessed.)
2. Your report is limited to four (4) pages; only the first four pages of your submission will be marked.
3. Submit your answer as an A4 PDF document.

4. Use a 12pt font as the primary font in your submission.
5. Often students feel that they simply *must* clarify some aspect in a way that cannot be done within four pages. If you must, you may include up to two pages of addenda beyond the initial four pages. They will not be marked. However, the lecturer has in some cases in the past briefly looked at addenda if some issue seemed particularly intriguing.
6. It is not foreseen that references will contribute to this assignment; if you wish to include references, they may spill over beyond page 4 to pages that form part of the addenda.
7. Double check that your name appears on your submission: The submissions will be printed to be marked and any identifying information in, for example, the file name, will be lost.
8. The assignments will be printed in monochrome to be marked — do not rely on colour.

Have fun!

References

- [1] M. S. Olivier. On the classification of questioned PDF documents — attributing PDF documents to the tools that created them. *Forensic Science International: Digital Investigation*, 56:302104, June 2026.